STIGMA Study End-of-Term Presentation:

Exploratory Analysis of Stress, Sleep, Activity and Recovery Across Menstrual Phases

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STIGMA Study Background

STIGMA - Addressing gaps in menstrual health research

Historical exclusion of Black and Brown communities in reproductive health research created lasting mistrust

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Previous wearables research studies have been small, homogenous, exclusionary (white, regularly cycling women only)

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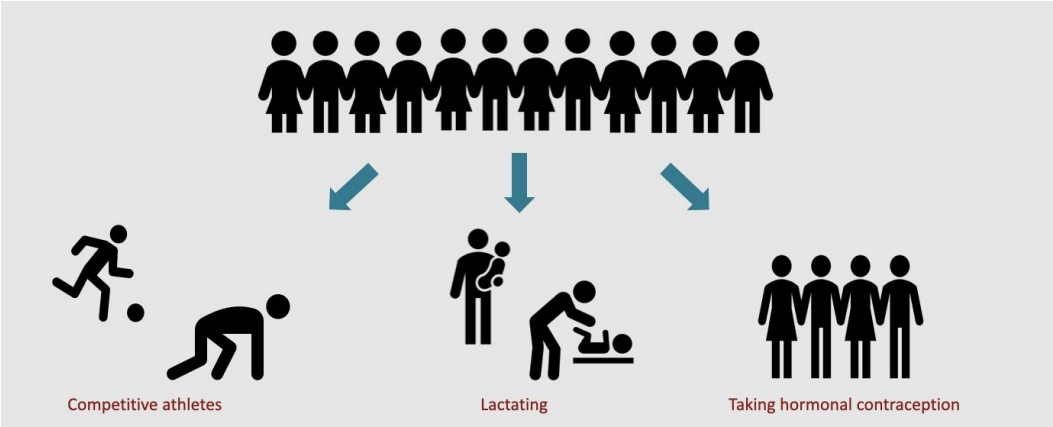
STIGMA: inclusive, participant-centered design to shift that narrative

Study Design

N = 304 participants total

Subcohorts:

- 40 competitive athletes
- 17 lactating
- 92 on hormonal contraception
- 221 naturally cycling



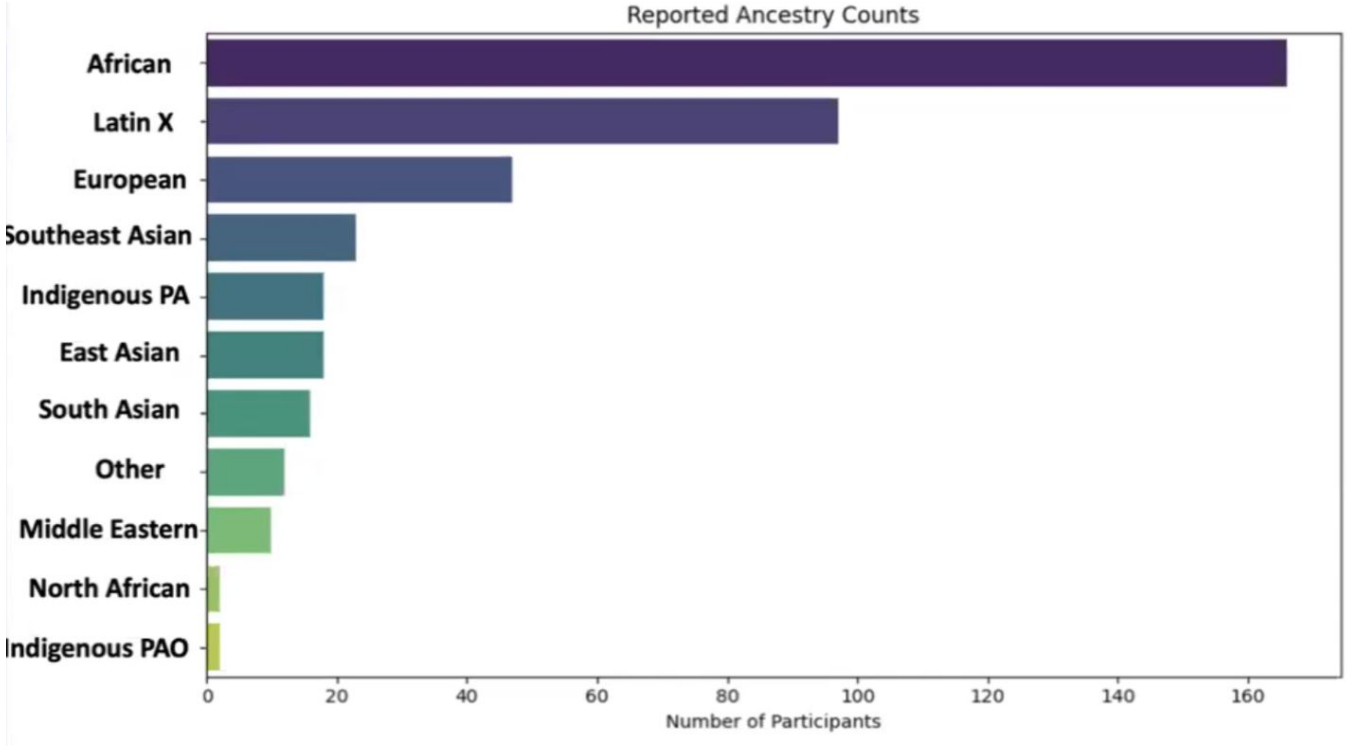
Recruitment: flyers, word of mouth, BIPOC-centered outreach

Consent: 30-min one-on-one sessions to explain goals, responsibilities, and participant voices

Participant Background

Age mean: 29 +- 7

Weight mean: 154 lbs +- 34



Participant Background

Age mean: 29 +- 7

Weight mean: 154 lbs +- 34

General Health Condition	N	Mental Health Condition	N	Reproductive Health Conditions	N
Sinus/ Seasonal allergies	57	Generalized Anxiety Disorder	64	Uterine Fibroids	24
Anemia	53	Attention-Deficit/Hyperactivity Disorder (ADHD)	44	Polycystic Ovarian Syndrome (PCOS)	23
Asthma	44	Clinical Depression	43	Frequent and/or persistent UTIs, BV, Yeast Infections	18
Headaches	30	Post Traumatic Stress Disorder (PTSD)	23	Ovarian cysts	16
Chronic inflammatory skin condition (e.g. eczema)	25	Attention-Deficit Disorder (ADD)	13	Premenstrual Syndrome (PMS)	14
Chronic pain (back, neck, hips, joint)	18	Postpartum Depression	13	Dysmenorrhea	10
Chronic migraines	16	Addiction	9	Endometriosis	8
Irritable Bowl Syndrome (IBS)	10	Obsessive Compulsive Disorder (OCD)	7	Amenorrhea	4
High Blood Pressure	9	Bipolar Disorder	4	Tilted Uterus	4
Insomnia	9	Premenstrual Dysphoric Disorder (PMDD)	3	Follicular cysts	2
Low Blood Pressure	7	Austism	1	Perimenopause	2
Seizures/Convulsions	3				
Type two diabetes	2				
Chest Pains	2				
Arthritis	2				
Neurological condition	1				
Heart Conditions	1				

Purpose and Research Goals

Educate, Empower, Execute

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Increase metabolic health knowledge in a personalized way

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Give people a sense of bodily autonomy

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Create healthier lifestyles

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Wearable technology is an excellent health care tool and research tool.

My Population

- Subset of STIGMA participants with valid cycle phases and Oura data
- Only participants with ≥3 ovulation dates
- Not all 304 participants, only those with overlapping stress, sleep, and activity data
- ~2300–2700 daily observations across valid participants
- Phase distribution: follicular and luteal dominate, ovulation smallest

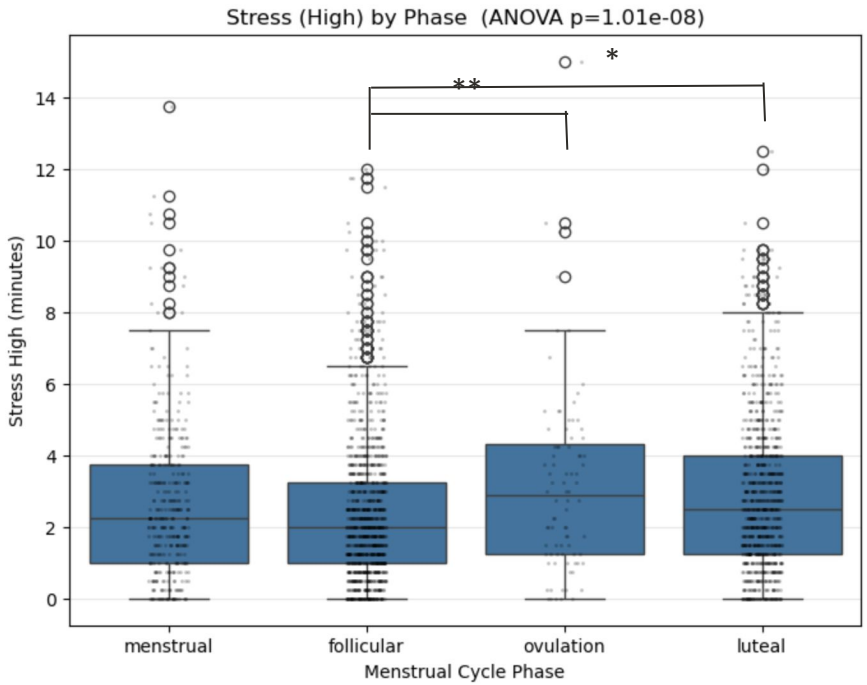
Single Variables

vs.

Menstrual Phase

Stress High by Phase

Stress High = amount of stressed time, calculated from stress sensor samples.



	stress_high		
	mean	std	count
phase			
follicular	2.345303	1.994329	1107
luteal	2.894072	2.164277	911
menstrual	2.670548	2.165426	365
ovulation	3.184524	2.627221	84

stress_high: η^2 = 0.016 (effect size)

Multiple Comparison of Means - Tukey HSD, FWER=0.05

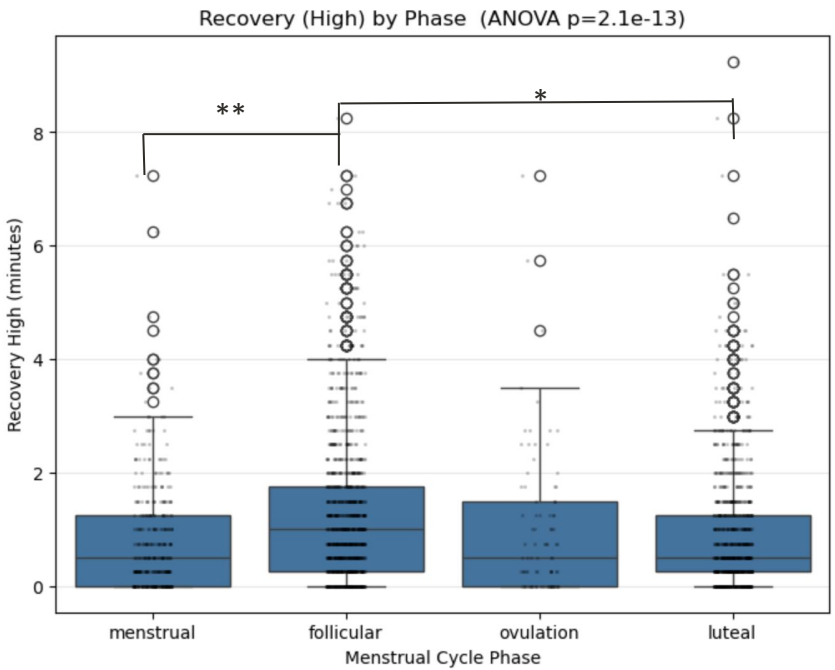
group1	group2	meandiff	p-adj	lower	upper	reject
follicular	luteal	0.5488	0.0	0.3064	0.7911	True
follicular	menstrual	0.3252	0.0519	-0.0018	0.6523	False
follicular	ovulation	0.8392	0.0025	0.226	1.4524	True
luteal	menstrual	-0.2235	0.3175	-0.5592	0.1121	False
luteal	ovulation	0.2905	0.6215	-0.3274	0.9083	False
menstrual	ovulation	0.514	0.1826	-0.1417	1.1697	False

ANOVA F=13.5, p=1.0e-08, η^2=0.016

follicular: n = 1107 luteal: n = 911 menstrual: n = 365 ovulation: n = 84
total n = 2467

Recovery High by Phase

Recovery High = amount of recovery time, calculated from stress sensor samples.



	recovery_high		
	mean	std	count
phase			
follicular	1.289747	1.345620	1107
luteal	0.928650	1.122931	911
menstrual	0.824658	1.004664	365
ovulation	1.035714	1.328637	84

Multiple Comparison of Means - Tukey HSD, FWER=0.05

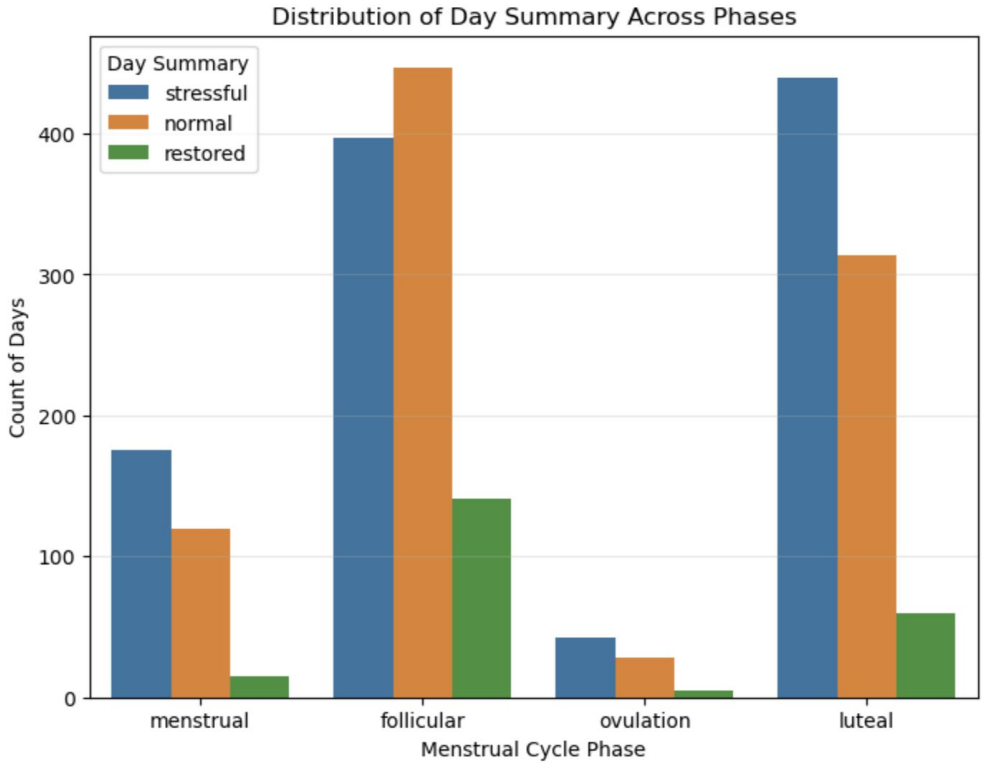
group1	group2	meandiff	p-adj	lower	upper	reject
follicular	luteal	-0.3611	0.0	-0.5014	-0.2208	True
follicular	menstrual	-0.4651	0.0	-0.6543	-0.2758	True
follicular	ovulation	-0.254	0.2546	-0.6089	0.1008	False
luteal	menstrual	-0.104	0.5143	-0.2982	0.0902	False
luteal	ovulation	0.1071	0.8681	-0.2505	0.4646	False
menstrual	ovulation	0.2111	0.4807	-0.1684	0.5905	False

ANOVA F=21.0, p=2.1e-13, η^2=0.025

follicular: n = 1107 luteal: n = 911 menstrual: n = 365 ovulation: n = 84
total n = 2467

Day Summary by Phase

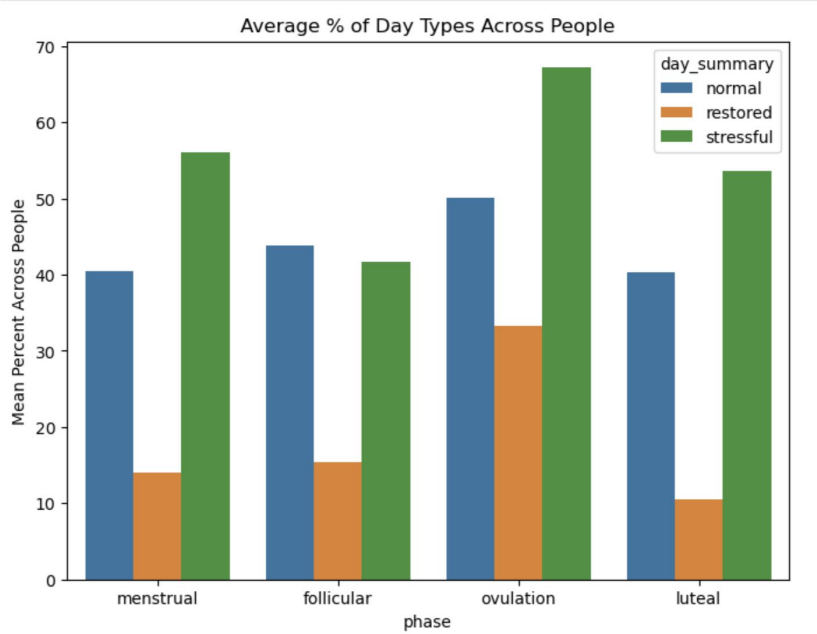
Day Summary = Categorical variable generated by Oura, classifying each day as Restored, Normal, or Stressful.

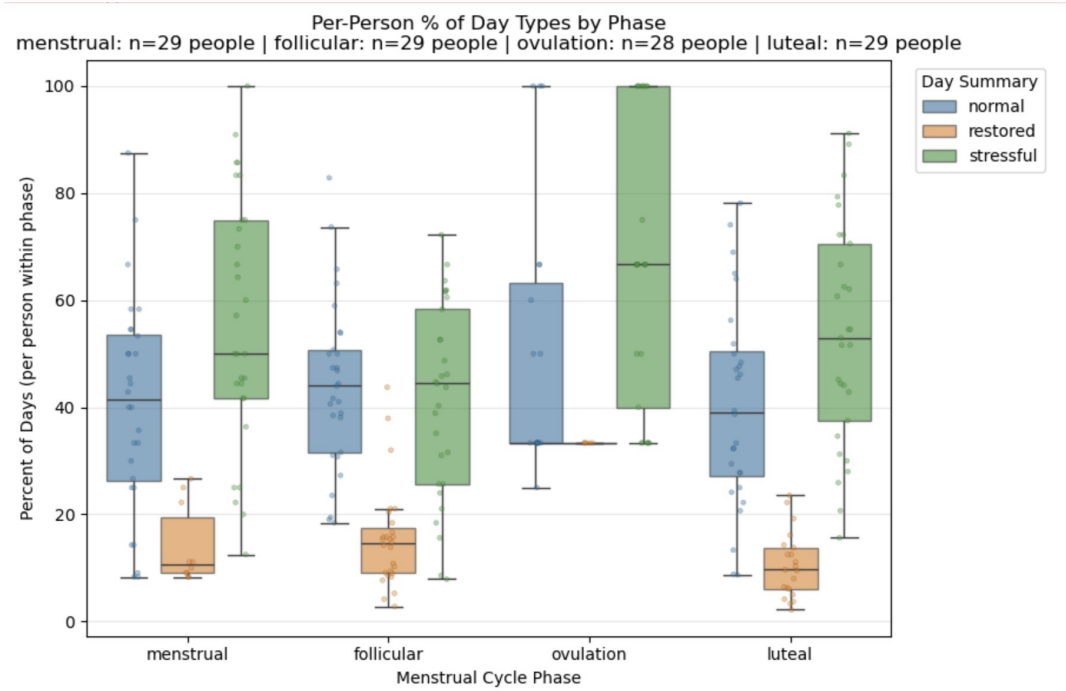


menstrual: n=310 follicular: n=985 ovulation: n=75 luteal: n=814

Day Summary by Phase

Day Summary = Categorical variable generated by Oura, classifying each day as Restored, Normal, or Stressful.

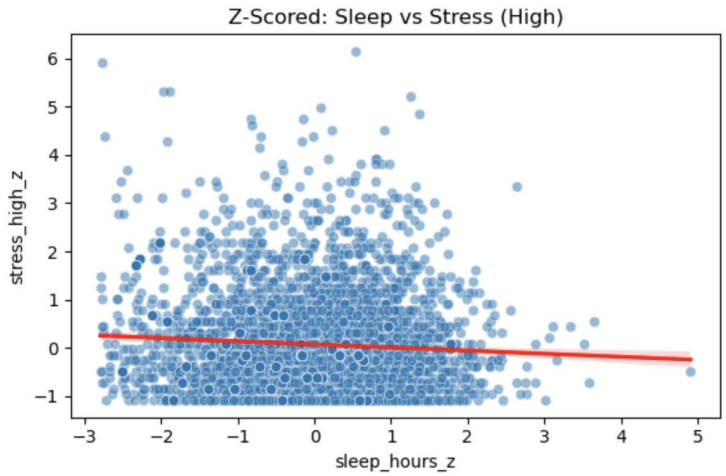




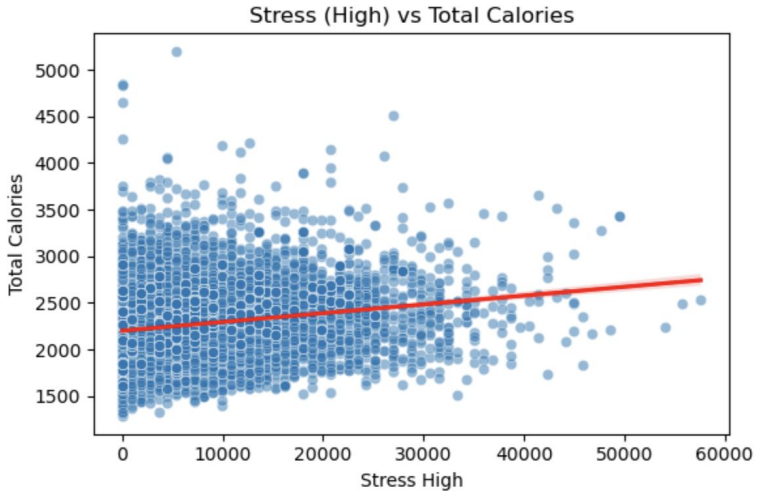
Variable Relationships vs. Menstrual Phase

Variable Trends

- Significant Trends:
  - Sleep vs Stress (r = -0.065, p = 3.89e-05)
  - Sleep vs Total Calories (r = -0.052, p = 0.00105)
  - Resting Time vs Recovery (r = 0.073, p = 2e-09)
  - High Activity vs Stress(r = 0.062, p = 3.99e-07)
  - High Activity vs Recovery (r = -0.094, p = 8.53e-15)
  - Stress vs Total Calories (r = 0.184, p = 1.7e-52)
- All have high significance (low p-value) and low correlation (r close to 0)



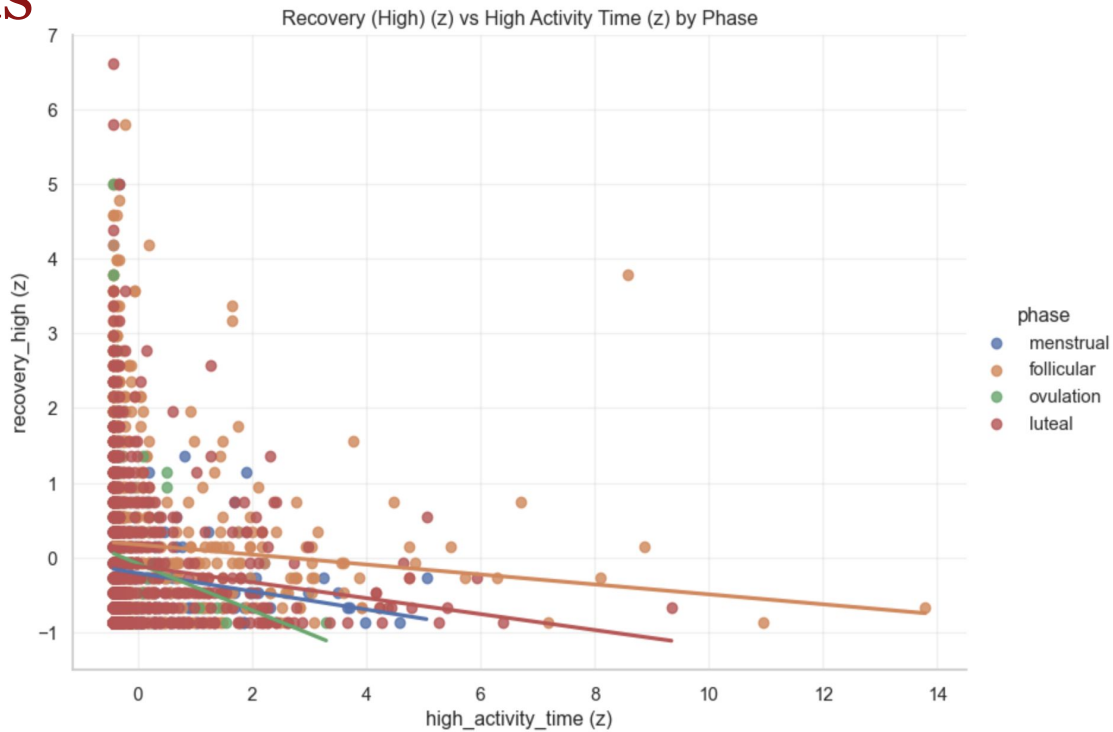
Z-Scored: Sleep vs Stress (High): r = -0.065, p = 3.89e-05, n = 3984



Stress (High) vs Total Calories: r = 0.184, p = 1.74e-52, n = 6766

Phase Interaction Tests

- Tested whether these trends change across menstrual cycle phases (interaction terms)
- Most interactions were not significant (phase doesn't strongly alter the correlations)
- Recovery vs High Activity Time showed a borderline interaction ($p \approx 0.07$).
- Suggests negative impact of high activity on recovery could be stronger in luteal & menstrual phases.

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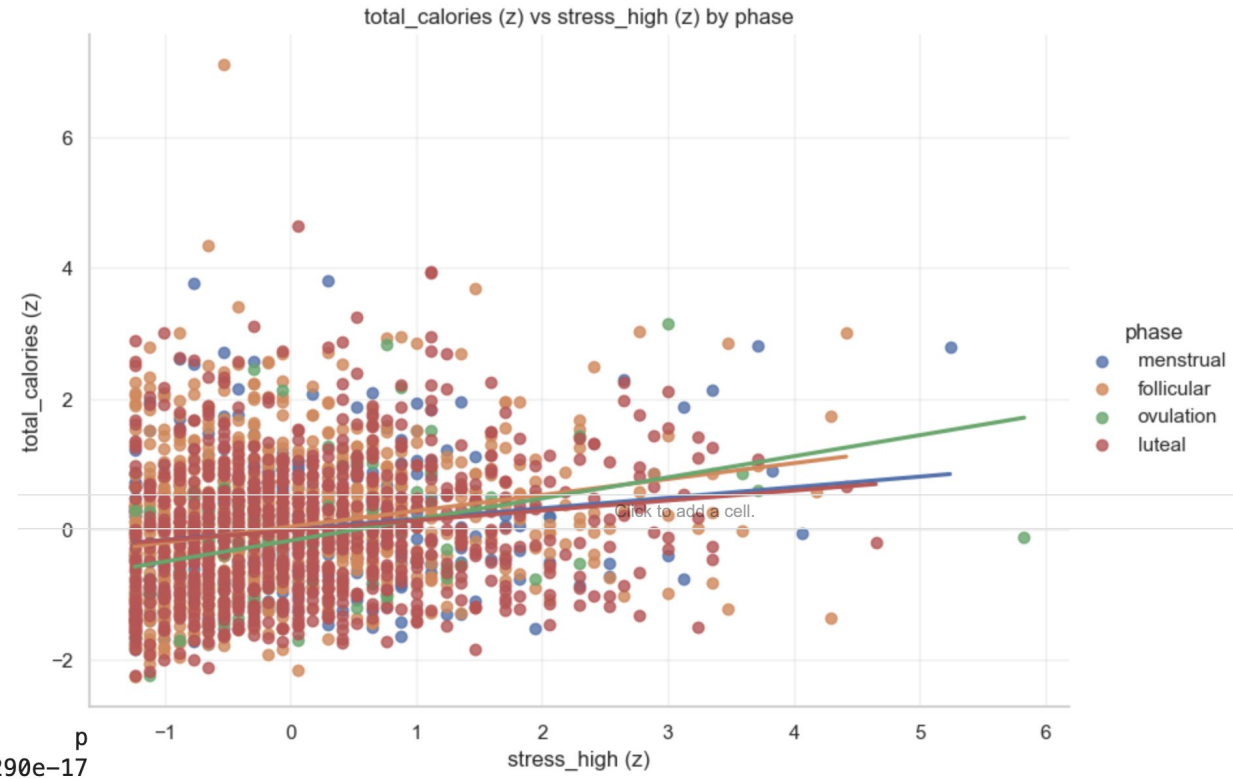
Per-phase slopes & correlations:
     phase    n     slope         r            p
follicular 1250 -0.004783 -0.074898 8.070536e-03
    luteal 1010 -0.010287 -0.179517 9.203119e-09
 menstrual  410 -0.011401 -0.175576 3.541535e-04
 ovulation   92 -0.017976 -0.178572 8.854828e-02
```

Stress vs Calories by Phase

- Stress vs calories had the strongest correlation ($r \approx 0.18$, $p < 1e\text{-}25$)
- Phase interaction is borderline ($p \approx 0.09$)
- Slopes:
  - Follicular/Ovulation - stronger positive slopes
  - Luteal - positive but smaller
  - Menstrual - weakest effect

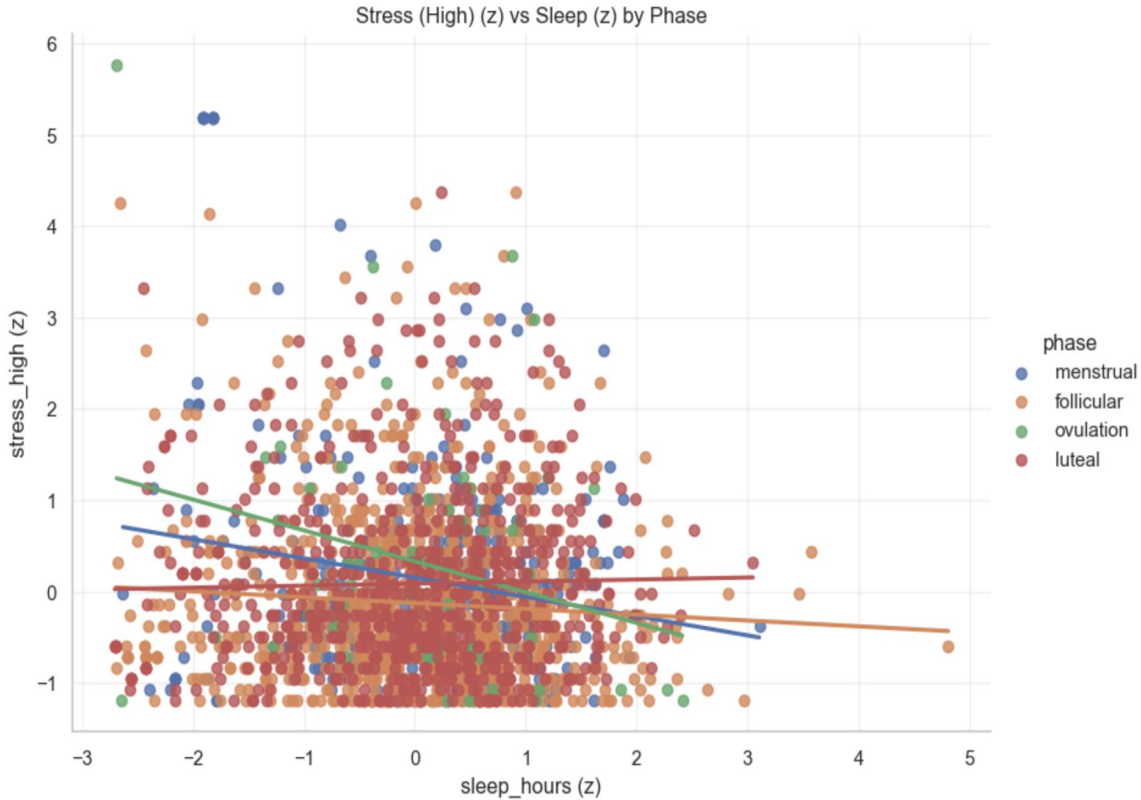


phase	n	slope	r	p
follicular	1250	0.778815	0.232028	9.662290e-17
luteal	1010	0.552355	0.174660	2.311679e-08
menstrual	410	0.417194	0.146589	2.927759e-03
ovulation	92	0.901860	0.336189	1.051226e-03)

Stress vs Sleep by Phase

- More sleep = less stress ($F = 16.97$, $p = 3.9e\text{-}05$).
- Interaction (phase):
  - Not significant ($p = 0.16$, $\Delta R^2 = 0.0022$).
- Per-phase slopes & correlations:

phase	n	slope	r	p
follicular	1144	-5.741697	-0.070791	0.016632
luteal	739	-6.795984	-0.080830	0.028006
menstrual	379	-9.088140	-0.092961	0.070653
ovulation	73	-29.541234	-0.253286	0.030610



Limitations and Next Steps

- Population: Get all participants phase dates so that this analysis can be done on the full population
- Effect sizes: All correlations were very small limiting usefulness for predictions, results can only be interpreted as trends not strong effects
- Next steps:
  - Test more variables and outcomes
  - Refine per-phase models
  - Integrate survey/symptom data
  - Explore mechanisms behind phase-specific differences

I want to see how these trends differ by individual participant and by specific groups:

Naturally cycling, PCOS NC, Copper IUD, Hormonal IUD, Oral BC, Perimenopausal, Athletes

Acknowledgements

STIGMA Team

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